2025/11/4

2026 年度　コンパイラ演習課題 (5)

- TACT の課題の添付ファイルとして PDF ファイルで提出すること.
- 手書きレポートを PDF として提出方法は講義 web サイトを参照[1]すること.
- 提出期限: 11 月 11 日 (火) 23:55(JST)

[**1**] 次の CF 文法 G_1 について答えよ.

$G_1 = \langle \{S, C, D, E\}, \{\texttt{do}, \texttt{while}, \texttt{(}, \texttt{)}, \texttt{;}, \texttt{=}, \texttt{if}, \texttt{else}, \texttt{\&\&}, \texttt{<}, \texttt{-}, \texttt{id}\}, P, S \rangle$

$$P = \left\{ \begin{array}{lll} 1: \ S \to \texttt{do}\ S^{(1)}\ \texttt{while (}E\texttt{);} & 4: \ E \to E^{(1)}\texttt{\&\&}D & 7: \ D \to C \\ 2: \ S \to \texttt{id=}E & 5: \ E \to D & 8: \ C \to C^{(1)}\texttt{-id} \\ 3: \ S \to \texttt{if(}E\texttt{)}\ S^{(1)}\ \texttt{else}\ S^{(2)} & 6: \ D \to D^{(1)}\texttt{<}C & 9: \ C \to \texttt{id} \end{array} \right\}$$

(a) 規則 1 のコード生成ルーチンを与えよ.

$$S \to \texttt{do}\{\boxed{\quad (1) \quad}\}\ S^{(1)}\texttt{while(}E\texttt{);}\{\boxed{\quad (2) \quad}\}$$

(b) その他の規則に対する生成ルーチンは以下のようになる. (裏面へ続く)

S	$\to$	$\texttt{ID=}E$	$\{\texttt{generate(`ID.}name\texttt{:=}E.var\texttt{')}\}$
S	$\to$	$\texttt{if(}E\texttt{)}$	$\{S.label_1\texttt{:=newlabel();generate(`if_not }E.var\texttt{ goto }S.label_1\texttt{ ')}\}$
		$S^{(1)}$	$\{S.label_2\texttt{:=newlabel();generate(`goto}S.lable_2\texttt{')}\}$
		else	$\{\texttt{generate(`label}S.label_1\texttt{')}\}$
		$S^{(2)}$	$\{\texttt{generate(`label}S.label_2\texttt{')}\}$
E	$\to$	$E^{(1)}\texttt{\&\&}D$	$\{E.var := \texttt{newTemp();}$
			$\texttt{generate(`}E.var\texttt{:=}E^{(1)}.var \ \wedge \ D.var\texttt{')}\}$
E	$\to$	D	$\{E.var := \texttt{newTemp();generate(`}E.var\texttt{:=}D.var\texttt{')}\}$
D	$\to$	$D^{(1)} < C$	$\{D.var := \texttt{newTemp();}$
			$\texttt{generate(`}D.var\texttt{:=}D^{(1)}.var < C.var\texttt{')}\}$
D	$\to$	C	$\{D.var := \texttt{newTemp();generate(`}D.var\texttt{:=}C.var\texttt{')}\}$
C	$\to$	$C^{(1)} - \texttt{ID}$	$\{C.var := \texttt{newTemp();}$
			$\texttt{generate(`}C.var\texttt{:=}C^{(1)}.var - \texttt{ID.}name\texttt{')}\}$
C	$\to$	ID	$C.var := \texttt{newTemp();generate(`}C.var\texttt{:=ID.}name\texttt{')}$

```
do if (x1<x2) x1=x1-x3 else x2=x2-x4 while(x5<x1-x6 && x7<x1);
```

から生成される 3 番地コードのプログラムを示せ.

[1] https://sites.google.com/sqlab.jp/26compiler/how-to-make-report